20/8/2025

Periodic Signals

$$x(t) = x(t+T) \quad \forall\, t \qquad \text{---------- (1)}$$

T = Smallest +ve constant for which the above eqn. holds true.

≡ fundamental time period

$x(t)$ is fn. of time t

$$\Rightarrow \boxed{x(t) = e^{j\omega t}}$$

$$x(t+T) = e^{j\omega(t+T)} = e^{j\omega t}\, e^{j\omega T}$$

$e^{j\omega t}$ is periodic with fundamental time $= \frac{2\pi}{\omega}$

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if Signal $x(t)$ is periodic $\forall\, t$

then $x(t) = x(t+T)$

$$\boxed{e^{j\omega t} = e^{j\omega t}\, e^{j\omega T}}$$

holds true iff $e^{j\omega T} = 1$

$\cos\omega T + j\sin\omega T = 1$

$\omega T = 2\pi k$

$$T = \frac{2\pi k}{\omega}$$

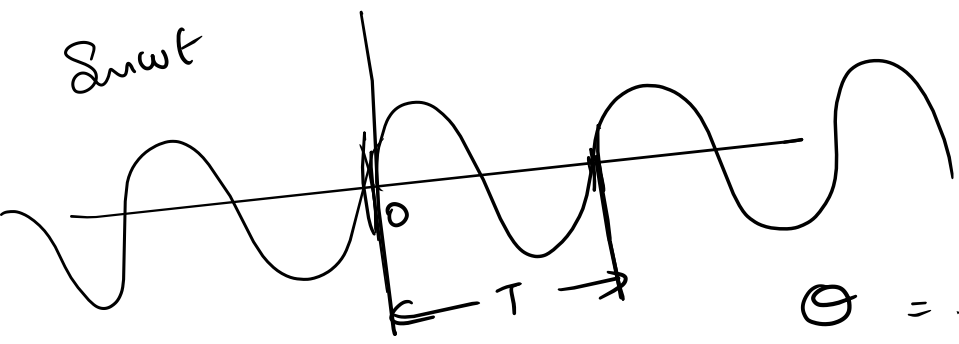
$k = 1$

$$T = \frac{2\pi}{\omega}$$

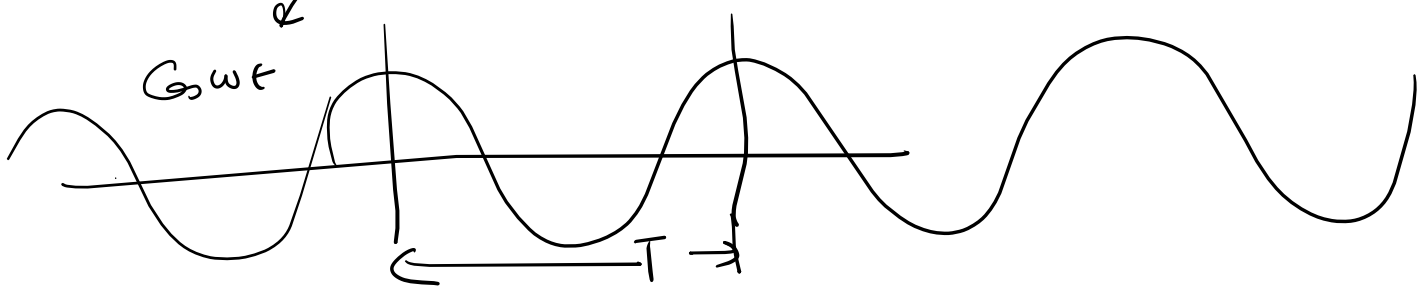
$x(t) = e^{j\omega t}$

$r = 1$

$\theta = \omega t$



$x(t) = \cos\omega t + j\sin\omega t$



$x(t) = e^{-j\omega t}$

$T = \frac{2\pi}{|\omega|}$

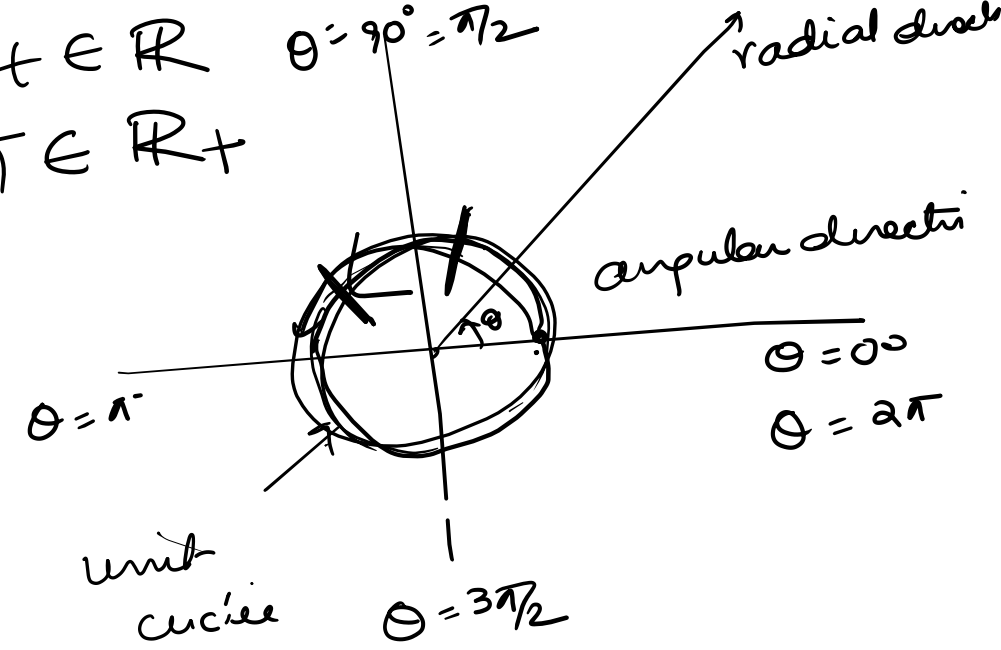
$r = 1$

$\theta = -\omega t$

$\theta = \omega t \Big|_{t=T} = 2\pi$

$t \in \mathbb{R}$

$T \in \mathbb{R}+$



$e^{j\omega t} = re^{j\theta}$

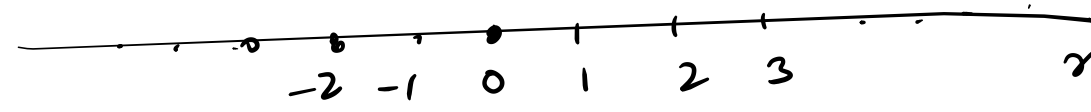
$r = 1$

$\theta = \omega t$

$E_\infty = \infty$

$P_\infty = 1$

$x[n] = e^{j\omega n}$ D.T. Signal



For Periodic Signal

$$x[n+N] = x[n] \quad \forall n$$

N = Smallest +ve integer

$N \in \mathbb{Z}^+$

$n \in \mathbb{Z}$

$$x[n+N] = e^{j\omega(n+N)} = e^{j\omega n} \cdot e^{j\omega N}$$

$$= e^{j\omega n}$$

$$= x[n]$$

if $e^{j\omega N} = 1$

$\omega N = 2\pi k$

$N = \frac{2\pi k}{\omega}$

N & k are integer

$\frac{2\pi}{\omega} = \frac{N}{k} \equiv$ rational no.

$$x_0[n] = e^{j\frac{24\pi n}{31}}$$

$$x_0[n+N]$$

$$= e^{j\frac{24\pi}{31}n} \cdot e^{j\frac{24\pi N}{31}}$$

$$\frac{24}{31}N = 2k$$

$$N = \frac{31}{12}k$$

For $k = 12$

$N = 31$ smallest

+ve integer

$$x_1[n] = e^{j\left(\frac{n}{6}\right)} = e^{j\omega n}$$

$$\Rightarrow \omega N = \frac{N}{6} = 2\pi k$$

$$\frac{N}{k} = 12\pi$$

Not an integer

Not periodic

$$x_2[n] = e^{j\frac{\pi n}{6}}$$

$$x_2[n+N] = e^{j\frac{\pi n}{6}} \cdot e^{j\frac{\pi N}{6}}$$

$$\frac{\pi N}{6} = 2\pi k$$

$$N = 12k$$

$\boxed{N = 12}$

periodic with fundamental time period $N = 12$

Concept of frequency

C.T. signals

$\omega = 2\pi f$

$x(t) = e^{j\omega t}$

$= e^{j\frac{2\pi}{T}t} = e^{j2\pi ft}$

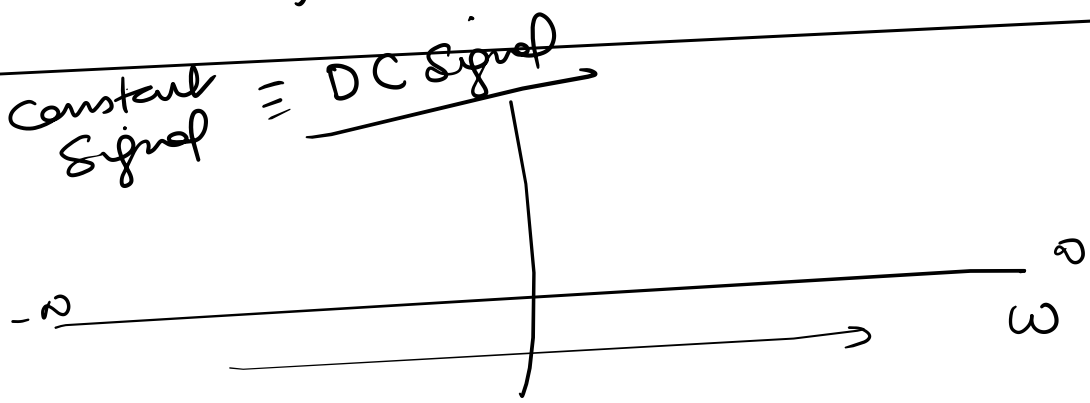
ω = radians/sec.

f = Hz $\equiv$ per sec.

$\omega = 0$

$x_1(t) = 1$

= constant signal $\equiv$ DC signal

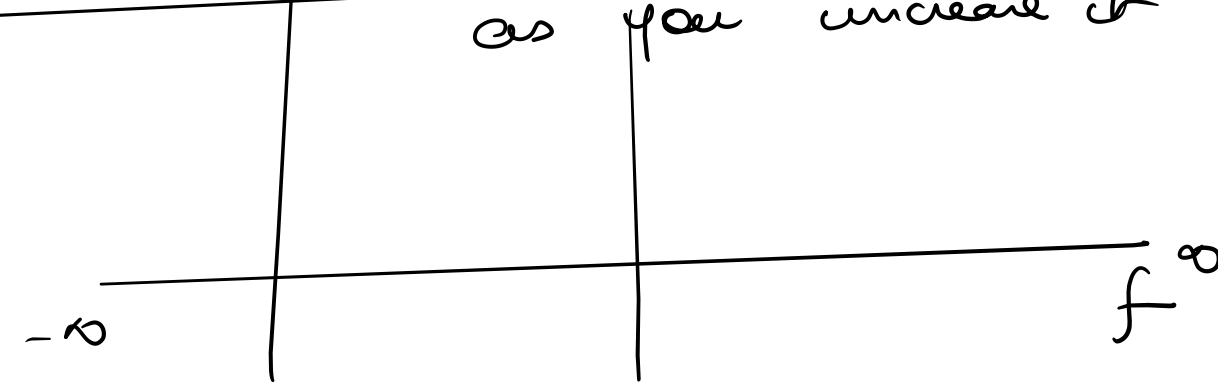


$x_2(t) = \cos \omega t$

$x_2(t) = \cos 2\pi f t$

$= \cos \frac{2\pi}{T} t$

frequencies are unique & keep on increasing as you increase it



D.T. Signal

$$x[n] = e^{j\omega n} = e^{j\frac{2\pi}{N}n} \qquad \omega = \frac{2\pi}{N}$$

$$k = 0, 1, 2 \ldots N-1$$

$$\phi_k[n] = e^{jk\frac{2\pi}{N}n}$$

N no. of signals have been defined

$$\phi_0[n] = 1 \qquad \phi_N[n] = e^{j\frac{N\cdot 2\pi}{N}n} = e^{j2\pi n} = 1$$

$$\phi_1[n] = e^{j\frac{2\pi}{N}n} \qquad \phi_{N+1}[n] = e^{j2\pi n}\, e^{j\frac{2\pi}{N}n} = e^{j\frac{2\pi}{N}n} = \phi_1[n]$$

$$\phi_2[n] = e^{j\frac{4\pi}{N}n}$$

$$\vdots$$

$$\phi_{N-1}[n] = e^{j\frac{2\pi}{N}(N-1)n} \qquad \phi_{2N-1}[n] = \phi_{N-1}[n]$$

$$\phi_{2N}[n] = \phi_0[n]$$

D.T. Signal

$x[n] = e^{j\omega n}$

$\omega = 0$

$= e^{j0.n} = e^{j0}$

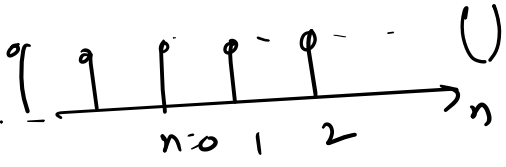
$= e$

$x(n) = e^{j\frac{5}{2}\pi n} = e^{j2\pi n} \cdot e^{j\frac{\pi}{2}n}$ $\quad \omega = \pi/2$

$= 1$

$x[n] = e^{j\frac{7}{3}\pi n} = e^{j2\pi n} \cdot e^{j\frac{\pi}{3}n}$ $\quad \omega = \pi/3$

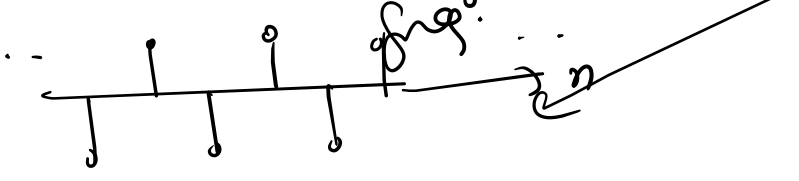
$= 1$



Unique frequencies are only between $0 \text{ —— } 2\pi$ interval

any continuous interval $-\pi \text{ —— } \pi$

of 2π



$e^{j0.n} = 1$ no variation

$e^{j\pi n} = (e^{j\pi})^n = 1, -1, 1, -1, \ldots\ldots$

Energy of D.T. signal

$$E_\infty = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

$$P_\infty = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$$

Ex

$$x[n] = \cos\frac{\pi}{4}n$$

$$\omega N = \frac{\pi}{4} N = 2\pi k$$

$$N = 8$$

$$E_N = \sum_{n=0}^{7} \left|\cos\frac{\pi}{4}n\right|^2 = \sum_{n=0}^{7} \cos^2\frac{\pi}{4}n$$

$$= \sum_{n=0}^{7} \left(\frac{1+\cos\frac{\pi}{2}n}{2}\right) = \frac{1}{2}\sum_{n=0}^{7} 1 + \frac{1}{2}\sum_{n=0}^{7}\cos\frac{\pi}{2}n$$

$$= \frac{1}{2}\times 8 + \frac{1}{2}\left[1+0+(-1)+0+1+0+(-1)+0\right]$$

$$= 4$$

$E_\infty = \infty$

$$P_\infty = \lim_{N\to\infty} \frac{1}{N} E_N = \lim_{N\to\infty} \frac{4}{8} = \frac{1}{2} = P_\infty$$